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TECHNIQUES FOR ENERGY-AWARE SCHEDULING IN WIRELESS COMMUNICATIONS

Abstract

Aspects described herein relate to a user equipment (UE) transmitting, for a network entity, an indication of available energy at the UE for a period of time and a buffer size at the UE, receiving, from the network entity and based on transmitting the indication of the available energy, a sequence of one or more scheduling grants indicating resources for the UE to use in transmitting the communications in the period of time, and allocating, based on the available energy and according to the sequence of one or more scheduling grants, the transmit power for transmitting the communications in the resources over the period of time or one or more additional periods of time. Other aspects relate to the network entity receiving the indication and transmitting the sequence of scheduling grants.

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Background/Summary

FIELD OF THE DISCLOSURE

[0001] Aspects of the present disclosure relate generally to wireless communication systems, and more particularly, to scheduling devices for wireless communications.

DESCRIPTION OF RELATED ART

[0002] Wireless communication systems are widely deployed to provide various types of communication content such as voice, video, packet data, messaging, broadcast, and so on. These systems may be multiple-access systems capable of supporting communication with multiple users by sharing the available system resources (e.g., time, frequency, and power). Examples of such multiple-access systems include code-division multiple access (CDMA) systems, time-division multiple access (TDMA) systems, frequency-division multiple access (FDMA) systems, and orthogonal frequency-division multiple access (OFDMA) systems, and single-carrier frequency division multiple access (SC-FDMA) systems.

[0003] These multiple access technologies have been adopted in various telecommunication standards to provide a common protocol that enables different wireless devices to communicate on a municipal, national, regional, and even global level. For example, a fifth generation (5G) wireless communications technology (which can be referred to as 5G new radio (5G NR)) is envisaged to expand and support diverse usage scenarios and applications with respect to current mobile network generations. In an aspect, 5G communications technology can include: enhanced mobile broadband addressing human-centric use cases for access to multimedia content, services and data; ultra-reliable-low latency communications (URLLC) with certain specifications for latency and reliability; and massive machine type communications, which can allow a very large number of connected devices and transmission of a relatively low volume of non-delay-sensitive information.

SUMMARY

[0004] The following presents a simplified summary of one or more aspects in order to provide a basic understanding of such aspects. This summary is not an extensive overview of all contemplated aspects, and is intended to neither identify key or critical elements of all aspects nor delineate the scope of any or all aspects. Its sole purpose is to present some concepts of one or more aspects in a simplified form as a prelude to the more detailed description that is presented later.

[0005] According to an aspect, an apparatus for wireless communication is provided that includes a transceiver, one or more memories configured to, individually or in combination, store instructions, and one or more processors communicatively coupled with the one or more memories. The one or more processors are, individually or in combination, configured to execute the instructions to cause the apparatus to transmit, for a network entity, an indication of available energy at the apparatus for a period of time and a buffer size at the apparatus, receive, from the network entity and based on transmitting the indication of the available energy, a sequence of one or more scheduling grants indicating resources for the apparatus to use in transmitting the communications in the period of time, and allocate, based on the available energy and according to the sequence of one or more scheduling grants, a transmit power for transmitting the communications in the resources over the period of time or one or more additional periods of time.

[0006] In another aspect, an apparatus for wireless communication is provided that includes a transceiver, one or more memories configured to, individually or in combination, store instructions, and one or more processors communicatively coupled with the one or more memories. The one or more processors are, individually or in combination, configured to execute the instructions to cause the apparatus to receive, for a user equipment (UE), an indication of available energy at the UE for a period of time and a buffer size at the UE, generate, for the UE and based on the available energy

and the buffer size at the UE, a scheduling decision including a target duty cycle for the UE, and transmit, for the UE and in accordance with the scheduling decision, a sequence of one or more scheduling grants indicating resources for the UE to use in transmitting the communications in the period of time.
[0007] In another aspect, a method for wireless communication at a UE is provided that includes transmitting, for a network entity, an indication of available energy at the UE for a period of time and a buffer size at the UE, receiving, from the network entity and based on transmitting the indication of the available energy, a sequence of one or more scheduling grants indicating resources for the UE to use in transmitting the communications in the period of time, and allocating, based on the available energy and according to the sequence of one or more scheduling grants, a transmit power for transmitting the communications in the resources over the period of time or one or more additional periods of time.
[0008] In another aspect, a method for wireless communication at a network node is provided that includes receiving, for a UE, an indication of available energy at the UE for a period of time and a buffer size at the UE, generating, for the UE and based on the available energy and the buffer size at the UE, a scheduling decision including a target duty cycle for the UE, and transmitting, for the UE and in accordance with the scheduling decision, a sequence of one or more scheduling grants indicating resources for the UE to use in transmitting the communications in the period of time.
[0009] In a further aspect, an apparatus for wireless communication is provided that includes a transceiver, a memory configured to store instructions, and one or more processors communicatively coupled with the transceiver and the memory. The one or more processors are configured to execute the instructions to perform the operations of methods described herein. In another aspect, an apparatus for wireless communication is provided that includes means for performing the operations of methods described herein. In yet another aspect, a computer-readable medium is provided including code executable by one or more processors to perform the operations of methods described herein.
[0010] To the accomplishment of the foregoing and related ends, the one or more aspects comprise the features hereinafter fully described and particularly pointed out in the claims. The following description and the annexed drawings set forth in detail certain illustrative features of the one or more aspects. These features are indicative, however, of but a few of the various ways in which the principles of various aspects may be employed, and this description is intended to include all such aspects and their equivalents.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0011] The disclosed aspects will hereinafter be described in conjunction with the appended drawings, provided to illustrate and not to limit the disclosed aspects, wherein like designations denote like elements, and in which:
[0012] FIG. **1** illustrates an example of a wireless communication system, in accordance with various aspects of the present disclosure;
[0013] FIG. **2** is a diagram illustrating an example of disaggregated base station architecture, in accordance with various aspects of the present disclosure;
[0014] FIG. **3** is a block diagram illustrating an example of a user equipment (UE), in accordance with various aspects of the present disclosure;
[0015] FIG. **4** is a block diagram illustrating an example of a base station, in accordance with various aspects of the present disclosure;
[0016] FIG. **5** illustrates an example of a scheduler architecture for a node scheduling a UE for wireless communications based on an indicated available energy at the UE, in accordance with

aspects described herein;

[0017] FIG. **6** illustrates an example of an outer scheduler architecture in a gNB, in accordance with aspects described herein;

[0018] FIG. **7** is a flow chart illustrating an example of a method for reporting available energy information for receiving a scheduling decision, in accordance with aspects described herein;

[0019] FIG. **8** is a flow chart illustrating an example of a method for scheduling a UE based on receiving available energy information regarding the UE, in accordance with aspects described herein; and

[0020] FIG. **9** is a block diagram illustrating an example of a MIMO communication system including a base station and a UE, in accordance with various aspects of the present disclosure.

DETAILED DESCRIPTION

[0021] Various aspects are now described with reference to the drawings. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of one or more aspects. It may be evident, however, that such aspect(s) may be practiced without these specific details.

[0022] The described features generally relate to indicating energy level information for the purpose of scheduling a device for wireless communications. For example, in fifth generation (5G) new radio (NR) or other wireless communication technologies, user equipment (UE) transmit power can be limited by regulatory constraints that impose restrictions on maximum permissible exposure by the UE, which can be measured or governed by specific absorption rate (SAR), e.g., for sub-6 gigahertz (GHz) bands, or power density (PD), e.g., for millimeter wave (mmW) bands. In 5G NR, impact of radio frequency (RF) exposure management for a UE is not conveyed to a network entity (e.g., a base station or gNB). Without the knowledge of UE energy availability at the gNB scheduler that schedules wireless communication resources for the UE, there may be a risk of allocating more resource blocks and higher transmit power than the UE can sustain and still comply with MPE requirements. This may lead to wasting cell resources and reducing the overall user experience. In accordance with aspects described herein, information related to RF exposure management (e.g., energy availability information) of a UE can be conveyed to the gNB scheduler for use in scheduling resources for the UE.

[0023] In 5G NR, regulatory constraints on RF exposure are determined by time-averaged RF exposure (and not by instantaneous RF exposure). A UE can determine the allowed RF exposure for a short duration of time into the future. The UE can translate this allowed RF exposure to instantaneous transmit power based on the transmission requirements in uplink. For example, the UE can average transmit power, as a proxy for exposure, over a time averaging window including a number of previous time divisions (e.g., slots) and one or more future time divisions. The UE can select a transmit power for the one or more future time divisions so that the average transmit power over the time averaging window, including the transmit power in the previous time divisions and the selected transmit power for the one or more future time divisions, complies with the SAR (or PD) limit.

[0024] In an example, additional signaling may allow the UE to report budget for RF exposure. For example, the UE may report energy headroom, which may be computed similar to the selected transmit power for the one or more future time divisions in the future. The UE may report the energy headroom per band in a carrier aggregation (CA) or dual/multiple connectivity configuration. In addition, for example, the UE can boost its power to support a subset of uplink grants that are received within a certain time window. For example, the UE can boost the power over the SAR (or PD) limit for one or more uplink grants and compensate the boost by decreasing the power for another one or more uplink grants, so the average transmit power over the set of uplink grants complies with the SAR (or PD) limit.

[0025] Typically, uplink schedulers (e.g., network entities, such as base stations, gNBs, etc. that schedule UEs with resources for uplink communications) can make decisions based on fairness

metrics (e.g., ensuring fairness in scheduling resources for multiple UEs), channel conditions, uplink buffer status, latency constraints, etc. A UE's ability to sustain high power transmissions for long periods of time (e.g., 100s of slots, 100-1000 ms duration) is typically not factored into scheduling decisions. Many attributes of uplink scheduling, however, may be impacted by a UE's ability to sustain high power transmissions, such as choice of rank, transmit precoding matrix indictor (TPMI), frequency of scheduling (also referred to as duty cycle), etc. Depending on energy availability per antenna, certain TPMIs may be able to sustain high power transmissions compared to others, certain ranks may be difficult to sustain, etc. In other words, in some examples, energy availability at a UE may be a metric that reflects a UE's ability to sustain high power transmissions.
[0026] In accordance with aspects described herein, available energy at a UE for a next time period and/or UE buffer size can be conveyed to the network entity scheduling the UE. In one example, the network entity can include a nested scheduler architecture where multiple nested schedulers can work at different time scales, and long-term, medium-term and/or near-term decisions can be handled by different nested entities making decisions appropriate for each time scale. For example, medium-term schedulers (e.g., loops on the order of 100s of milliseconds (ms)) may be most appropriate to handle energy availability at the UE. In any case, if the network entity (e.g., gNB) scheduling the UE is aware of the UE's energy budget, the network entity can adopt a more nuanced scheduling policy for scheduling resources for the UE. The network entity may optimize uplink duty cycle so that UE can sustain uplink transmissions at a high power throughout a certain time window. When the UE is able to sustain a high power, this can lead to an increase in spectral efficiency, data transfer efficiency, etc., which may lead to reduction in grants required to transfer the same amount of payload.
[0027] The described features will be presented in more detail below with reference to FIGS. **1**-**9**.
[0028] As used in this application, the terms “component,” “module,” “system” and the like are intended to include a computer-related entity, such as but not limited to hardware, firmware, a combination of hardware and software, software, or software in execution. For example, a component may be, but is not limited to being, a process running on a processor, a processor, an object, an executable, a thread of execution, a program, and/or a computer. By way of illustration, both an application running on a computing device and the computing device can be a component. One or more components can reside within a process and/or thread of execution and a component can be localized on one computer and/or distributed between two or more computers. In addition, these components can execute from various computer readable media having various data structures stored thereon. The components can communicate by way of local and/or remote processes such as in accordance with a signal having one or more data packets, such as data from one component interacting with another component in a local system, distributed system, and/or across a network such as the Internet with other systems by way of the signal.
[0029] As used herein, a processor, at least one processor, and/or one or more processors, individually or in combination, configured to perform or operable for performing a plurality of actions is meant to include at least two different processors able to perform different, overlapping or non-overlapping subsets of the plurality actions, or a single processor able to perform all of the plurality of actions. In one non-limiting example of multiple processors being able to perform different ones of the plurality of actions in combination, a description of a processor, at least one processor, and/or one or more processors configured or operable to perform actions X, Y, and Z may include at least a first processor configured or operable to perform a first subset of X, Y, and Z (e.g., to perform X) and at least a second processor configured or operable to perform a second subset of X, Y, and Z (e.g., to perform Y and Z). Alternatively, a first processor, a second processor, and a third processor may be respectively configured or operable to perform a respective one of actions X, Y, and Z. It should be understood that any combination of one or more processors each may be configured or operable to perform any one or any combination of a plurality of actions.
[0030] As used herein, a memory, at least one memory, and/or one or more memories, individually

or in combination, configured to store or having stored thereon instructions executable by one or more processors for performing a plurality of actions is meant to include at least two different memories able to store different, overlapping or non-overlapping subsets of the instructions for performing different, overlapping or non-overlapping subsets of the plurality actions, or a single memory able to store the instructions for performing all of the plurality of actions. In one non-limiting example of one or more memories, individually or in combination, being able to store different subsets of the instructions for performing different ones of the plurality of actions, a description of a memory, at least one memory, and/or one or more memories configured or operable to store or having stored thereon instructions for performing actions X, Y, and Z may include at least a first memory configured or operable to store or having stored thereon a first subset of instructions for performing a first subset of X, Y, and Z (e.g., instructions to perform X) and at least a second memory configured or operable to store or having stored thereon a second subset of instructions for performing a second subset of X, Y, and Z (e.g., instructions to perform Y and Z). Alternatively, a first memory, and second memory, and a third memory may be respectively configured to store or have stored thereon a respective one of a first subset of instructions for performing X, a second subset of instruction for performing Y, and a third subset of instructions for performing Z. It should be understood that any combination of one or more memories each may be configured or operable to store or have stored thereon any one or any combination of instructions executable by one or more processors to perform any one or any combination of a plurality of actions. Moreover, one or more processors may each be coupled to at least one of the one or more memories and configured or operable to execute the instructions to perform the plurality of actions. For instance, in the above non-limiting example of the different subset of instructions for performing actions X, Y, and Z, a first processor may be coupled to a first memory storing instructions for performing action X, and at least a second processor may be coupled to at least a second memory storing instructions for performing actions Y and Z, and the first processor and the second processor may, in combination, execute the respective subset of instructions to accomplish performing actions X, Y, and Z. Alternatively, three processors may access one of three different memories each storing one of instructions for performing X, Y, or Z, and the three processor may in combination execute the respective subset of instruction to accomplish performing actions X, Y, and Z. Alternatively, a single processor may execute the instructions stored on a single memory, or distributed across multiple memories, to accomplish performing actions X, Y, and Z.

[0031] Techniques described herein may be used for various wireless communication systems such as CDMA, TDMA, FDMA, OFDMA, single carrier-FDMA, and other systems. The terms “system” and “network” may often be used interchangeably. A CDMA system may implement a radio technology such as CDMA2000, Universal Terrestrial Radio Access (UTRA), etc. CDMA2000 covers IS-2000, IS-95, and IS-856 standards. IS-2000 Releases 0 and A are commonly referred to as CDMA2000 1×, 1×, etc. IS-856 (TIA-856) is commonly referred to as CDMA2000 1×EV-DO, High Rate Packet Data (HRPD), etc. UTRA includes Wideband CDMA (WCDMA) and other variants of CDMA. A TDMA system may implement a radio technology such as Global System for Mobile Communications (GSM). An OFDMA system may implement a radio technology such as Ultra Mobile Broadband (UMB), Evolved UTRA (E-UTRA), IEEE 802.11 (Wi-Fi), IEEE 802.16 (WiMAX), IEEE 802.20, Flash-OFDM™, etc. UTRA and E-UTRA are part of Universal Mobile Telecommunication System (UMTS). 3GPP Long Term Evolution (LTE) and LTE-Advanced (LTE-A) are new releases of UMTS that use E-UTRA. UTRA, E-UTRA, UMTS, LTE, LTE-A, and GSM are described in documents from an organization named “3rd Generation Partnership Project” (3GPP). CDMA2000 and UMB are described in documents from an organization named “3rd Generation Partnership Project 2” (3GPP2). The techniques described herein may be used for the systems and radio technologies mentioned above as well as other systems and radio technologies, including cellular (e.g., LTE) communications over a shared radio frequency spectrum band. The description below, however, describes an LTE/LTE-A system for purposes of example, and LTE terminology is used in much of the description below, although the techniques are applicable beyond LTE/LTE-A applications (e.g., to fifth generation (5G) new radio (NR) networks or other next generation communication systems).

[0032] The following description provides examples, and is not limiting of the scope, applicability, or examples set forth in the claims. Changes may be made in the function and arrangement of elements discussed without departing from the scope of the disclosure. Various examples may omit, substitute, or add various procedures or components as appropriate. For instance, the methods described may be performed in an order different from that described, and various steps may be added, omitted, or combined. Also, features described with respect to some examples may be combined in other examples.

[0033] Various aspects or features will be presented in terms of systems that can include a number of devices, components, modules, and the like. It is to be understood and appreciated that the various systems can include additional devices, components, modules, etc. and/or may not include all of the devices, components, modules etc. discussed in connection with the figures. A combination of these approaches can also be used.

[0034] FIG. **1** is a diagram illustrating an example of a wireless communications system and an access network **100**. The wireless communications system (also referred to as a wireless wide area network (WWAN)) can include base stations **102**, UEs **104**, an Evolved Packet Core (EPC) **160**, and/or a 5G Core (5GC) **190**. The base stations **102** may include macro cells (high power cellular base station) and/or small cells (low power cellular base station). The macro cells can include base stations. The small cells can include femtocells, picocells, and microcells. In an example, the base stations **102** may also include gNBs **180**, as described further herein. In one example, some nodes of the wireless communication system may have a modem **340** and UE communicating component **342** for reporting available energy information to a network entity for receiving an associated scheduling decision, in accordance with aspects described herein.

[0035] In addition, some nodes may have a modem **440** and BS communicating component **442** for scheduling resources for a UE based on received available energy information for the UE, in accordance with aspects described herein. Though a UE **104** is shown as having the modem **340** and UE communicating component **342** and a base station **102**/gNB **180** is shown as having the modem **440** and BS communicating component **442**, this is one illustrative example, and substantially any node or type of node may include a modem **340** and UE communicating component **342** and/or a modem **440** and BS communicating component **442** for providing corresponding functionalities described herein.

[0036] The base stations **102** configured for 4G LTE (which can collectively be referred to as Evolved Universal Mobile Telecommunications System (UMTS) Terrestrial Radio Access Network (E-UTRAN)) may interface with the EPC **160** through backhaul links **132** (e.g., using an S1 interface). The base stations **102** configured for 5G NR (which can collectively be referred to as Next Generation RAN (NG-RAN)) may interface with 5GC **190** through backhaul links **184**. In addition to other functions, the base stations **102** may perform one or more of the following functions: transfer of user data, radio channel ciphering and deciphering, integrity protection, head compression, mobility control functions (e.g., handover, dual connectivity), inter-cell interference coordination, connection setup and release, load balancing, distribution for non-access stratum (NAS) messages, NAS node selection, synchronization, radio access network (RAN) sharing, multimedia broadcast multicast service (MBMS), subscriber and equipment trace, RAN information management (RIM), paging, positioning, and delivery of warning messages. The base stations **102** may communicate directly or indirectly (e.g., through the EPC **160** or 5GC **190**) with each other over backhaul links **134** (e.g., using an X2 interface). The backhaul links **134** may be wired or wireless.

[0037] The base stations **102** may wirelessly communicate with one or more UEs **104**. Each of the base stations **102** may provide communication coverage for a respective geographic coverage area

110. There may be overlapping geographic coverage areas **110**. For example, the small cell **102′** may have a coverage area **110′** that overlaps the coverage area **110** of one or more macro base stations **102**. A network that includes both small cell and macro cells may be referred to as a heterogeneous network. A heterogeneous network may also include Home Evolved Node Bs (eNBs) (HeNBs), which may provide service to a restricted group, which can be referred to as a closed subscriber group (CSG). The communication links **120** between the base stations **102** and the UEs **104** may include uplink (UL) (also referred to as reverse link) transmissions from a UE **104** to a base station **102** and/or downlink (DL) (also referred to as forward link) transmissions from a base station **102** to a UE **104**. The communication links **120** may use multiple-input and multiple-output (MIMO) antenna technology, including spatial multiplexing, beamforming, and/or transmit diversity. The communication links may be through one or more carriers. The base stations **102**/UEs **104** may use spectrum up to Y MHz (e.g., 5, 10, 15, 20, 100, 400, etc. MHz) bandwidth per carrier allocated in a carrier aggregation of up to a total of Yx MHz (e.g., for x component carriers) used for transmission in the DL and/or the UL direction. The carriers may or may not be adjacent to each other. Allocation of carriers may be asymmetric with respect to DL and UL (e.g., more or less carriers may be allocated for DL than for UL). The component carriers may include a primary component carrier and one or more secondary component carriers. A primary component carrier may be referred to as a primary cell (PCell) and a secondary component carrier may be referred to as a secondary cell (SCell).

[0038] In another example, certain UEs **104** may communicate with each other using device-to-device (D2D) communication link **158**. The D2D communication link **158** may use the DL/UL WWAN spectrum. The D2D communication link **158** may use one or more sidelink channels, such as a physical sidelink broadcast channel (PSBCH), a physical sidelink discovery channel (PSDCH), a physical sidelink shared channel (PSSCH), and a physical sidelink control channel (PSCCH). D2D communication may be through a variety of wireless D2D communications systems, such as for example, FlashLinQ, WiMedia, Bluetooth, ZigBee, Wi-Fi based on the IEEE 802.11 standard, LTE, or NR.

[0039] The wireless communications system may further include a Wi-Fi access point (AP) **150** in communication with Wi-Fi stations (STAs) **152** via communication links **154** in a 5 GHz unlicensed frequency spectrum. When communicating in an unlicensed frequency spectrum, the STAs **152**/AP **150** may perform a clear channel assessment (CCA) prior to communicating in order to determine whether the channel is available.

[0040] The small cell **102′** may operate in a licensed and/or an unlicensed frequency spectrum. When operating in an unlicensed frequency spectrum, the small cell **102′** may employ NR and use the same 5 GHz unlicensed frequency spectrum as used by the Wi-Fi AP **150**. The small cell **102′**, employing NR in an unlicensed frequency spectrum, may boost coverage to and/or increase capacity of the access network.

[0041] A base station **102**, whether a small cell **102′** or a large cell (e.g., macro base station), may include an eNB, gNodeB (gNB), or other type of base station. Some base stations, such as gNB **180** may operate in a traditional sub 6 GHz spectrum, in millimeter wave (mmW) frequencies, and/or near mmW frequencies in communication with the UE **104**. When the gNB **180** operates in mmW or near mmW frequencies, the gNB **180** may be referred to as an mmW base station. Extremely high frequency (EHF) is part of the RF in the electromagnetic spectrum. EHF has a range of 30 GHz to 300 GHz and a wavelength between 1 millimeter and 10 millimeters. Radio waves in the band may be referred to as a millimeter wave. Near mmW may extend down to a frequency of 3 GHz with a wavelength of 100 millimeters. The super high frequency (SHF) band extends between 3 GHz and 30 GHz, also referred to as centimeter wave. Communications using the mmW/near mmW radio frequency band has extremely high path loss and a short range. The mmW base station **180** may utilize beamforming **182** with the UE **104** to compensate for the extremely high path loss and short range. A base station **102** referred to herein can include a gNB **180**.

[0042] The EPC **160** may include a Mobility Management Entity (MME) **162**, other MMEs **164**, a Serving Gateway **166**, a Multimedia Broadcast Multicast Service (MBMS) Gateway **168**, a Broadcast Multicast Service Center (BM-SC) **170**, and a Packet Data Network (PDN) Gateway **172**. The MME **162** may be in communication with a Home Subscriber Server (HSS) **174**. The MME **162** is the control node that processes the signaling between the UEs **104** and the EPC **160**. Generally, the MME **162** provides bearer and connection management. All user Internet protocol (IP) packets are transferred through the Serving Gateway **166**, which itself is connected to the PDN Gateway **172**. The PDN Gateway **172** provides UE IP address allocation as well as other functions. The PDN Gateway **172** and the BM-SC **170** are connected to the IP Services **176**. The IP Services **176** may include the Internet, an intranet, an IP Multimedia Subsystem (IMS), a PS Streaming Service, and/or other IP services. The BM-SC **170** may provide functions for MBMS user service provisioning and delivery. The BM-SC **170** may serve as an entry point for content provider MBMS transmission, may be used to authorize and initiate MBMS Bearer Services within a public land mobile network (PLMN), and may be used to schedule MBMS transmissions. The MBMS Gateway **168** may be used to distribute MBMS traffic to the base stations **102** belonging to a Multicast Broadcast Single Frequency Network (MBSFN) area broadcasting a particular service, and may be responsible for session management (start/stop) and for collecting eMBMS related charging information.
[0043] The 5GC **190** may include an Access and Mobility Management Function (AMF) **192**, other AMFs **193**, a Session Management Function (SMF) **194**, and a User Plane Function (UPF) **195**. The AMF **192** may be in communication with a Unified Data Management (UDM) **196**. The AMF **192** can be a control node that processes the signaling between the UEs **104** and the 5GC **190**. Generally, the AMF **192** can provide QoS flow and session management. User Internet protocol (IP) packets (e.g., from one or more UEs **104**) can be transferred through the UPF **195**. The UPF **195** can provide UE IP address allocation for one or more UEs, as well as other functions. The UPF **195** is connected to the IP Services **197**. The IP Services **197** may include the Internet, an intranet, an IP Multimedia Subsystem (IMS), a PS Streaming Service, and/or other IP services.
[0044] The base station may also be referred to as a gNB, Node B, evolved Node B (eNB), an access point, a base transceiver station, a radio base station, a radio transceiver, a transceiver function, a basic service set (BSS), an extended service set (ESS), a transmit reception point (TRP), or some other suitable terminology. The base station **102** provides an access point to the EPC **160** or 5GC **190** for a UE **104**. Examples of UEs **104** include a cellular phone, a smart phone, a session initiation protocol (SIP) phone, a laptop, a personal digital assistant (PDA), a satellite radio, a global positioning system, a multimedia device, a video device, a digital audio player (e.g., MP3 player), a camera, a game console, a tablet, a smart device, a wearable device, a vehicle, an electric meter, a gas pump, a large or small kitchen appliance, a healthcare device, an implant, a sensor/actuator, a display, or any other similar functioning device. Some of the UEs **104** may be referred to as IoT devices (e.g., parking meter, gas pump, toaster, vehicles, heart monitor, etc.). IoT UEs may include machine type communication (MTC)/enhanced MTC (eMTC, also referred to as category (CAT)-M, Cat M1) UEs, NB-IoT (also referred to as CAT NB1) UEs, as well as other types of UEs. In the present disclosure, eMTC and NB-IoT may refer to future technologies that may evolve from or may be based on these technologies. For example, eMTC may include FeMTC (further eMTC), eFeMTC (enhanced further eMTC), mMTC (massive MTC), etc., and NB-IoT may include eNB-IoT (enhanced NB-IoT), FeNB-IoT (further enhanced NB-IoT), etc. The UE **104** may also be referred to as a station, a mobile station, a subscriber station, a mobile unit, a subscriber unit, a wireless unit, a remote unit, a mobile device, a wireless device, a wireless communications device, a remote device, a mobile subscriber station, an access terminal, a mobile terminal, a wireless terminal, a remote terminal, a handset, a user agent, a mobile client, a client, or some other suitable terminology.
[0045] Deployment of communication systems, such as 5G new radio (NR) systems, may be

arranged in multiple manners with various components or constituent parts. In a 5G NR system, or network, a network node, a network entity, a mobility element of a network, a radio access network (RAN) node, a core network node, a network element, or a network equipment, such as a base station (BS, e.g., BS **102**), or one or more units (or one or more components) performing base station functionality, may be implemented in an aggregated or disaggregated architecture. For example, a BS (such as a Node B (NB), evolved NB (eNB), NR BS, 5G NB, access point (AP), a transmit receive point (TRP), or a cell, etc.) may be implemented as an aggregated base station (also known as a standalone BS or a monolithic BS) or a disaggregated base station.

[0046] An aggregated base station may be configured to utilize a radio protocol stack that is physically or logically integrated within a single RAN node. A disaggregated base station may be configured to utilize a protocol stack that is physically or logically distributed among two or more units (such as one or more central or centralized units (CUs), one or more distributed units (DUs), or one or more radio units (RUs)). In some aspects, a CU may be implemented within a RAN node, and one or more DUs may be co-located with the CU, or alternatively, may be geographically or virtually distributed throughout one or multiple other RAN nodes. The DUs may be implemented to communicate with one or more RUs. Each of the CU, DU and RU also can be implemented as virtual units, i.e., a virtual central unit (VCU), a virtual distributed unit (VDU), or a virtual radio unit (VRU).

[0047] Base station-type operation or network design may consider aggregation characteristics of base station functionality. For example, disaggregated base stations may be utilized in an integrated access backhaul (IAB) network, an open radio access network (O-RAN (such as the network configuration sponsored by the O-RAN Alliance)), or a virtualized radio access network (vRAN, also known as a cloud radio access network (C-RAN)). Disaggregation may include distributing functionality across two or more units at various physical locations, as well as distributing functionality for at least one unit virtually, which can enable flexibility in network design. The various units of the disaggregated base station, or disaggregated RAN architecture, can be configured for wired or wireless communication with at least one other unit.

[0048] In an example, UE communicating component **342** of a UE **104** can indicate, to a scheduling entity (e.g., a base station **102**, gNB **108**, sidelink UE, etc.) available energy information at the UE **104**. As described above and in further detail herein, BS communicating component **442** of a base station **102** or gNB **180**, etc., can generate a scheduling decision for the UE **104** based at least in part on the available energy information and can accordingly transmit a sequence of one or more scheduling grants to the UE **104**. The UE **104** can accordingly allocate transmit power for transmitting uplink communications over resources indicated in the sequence of one or more scheduling grants, as described herein.

[0049] FIG. **2** shows a diagram illustrating an example of disaggregated base station **200** architecture. The disaggregated base station **200** architecture may include one or more central units (CUs) **210** that can communicate directly with a core network **220** via a backhaul link, or indirectly with the core network **220** through one or more disaggregated base station units (such as a Near-Real Time (Near-RT) RAN Intelligent Controller (RIC) **225** via an E2 link, or a Non-Real Time (Non-RT) RIC **215** associated with a Service Management and Orchestration (SMO) Framework **205**, or both). A CU **210** may communicate with one or more distributed units (DUs) **230** via respective midhaul links, such as an F1 interface. The DUs **230** may communicate with one or more radio units (RUs) **240** via respective fronthaul links. The RUs **240** may communicate with respective UEs **104** via one or more radio frequency (RF) access links. In some implementations, the UE **104** may be simultaneously served by multiple RUs **240**.

[0050] Each of the units, e.g., the CUs **210**, the DUs **230**, the RUs **240**, as well as the Near-RT RICs **225**, the Non-RT RICs **215** and the SMO Framework **205**, may include one or more interfaces or be coupled to one or more interfaces configured to receive or transmit signals, data, or information (collectively, signals) via a wired or wireless transmission medium. Each of the units, or an associated processor or controller providing instructions to the communication interfaces of the units, can be configured to communicate with one or more of the other units via the transmission medium. For example, the units can include a wired interface configured to receive or transmit signals over a wired transmission medium to one or more of the other units. Additionally, the units can include a wireless interface, which may include a receiver, a transmitter or transceiver (such as a radio frequency (RF) transceiver), configured to receive or transmit signals, or both, over a wireless transmission medium to one or more of the other units.

[0051] In some aspects, the CU **210** may host one or more higher layer control functions. Such control functions can include radio resource control (RRC), packet data convergence protocol (PDCP), service data adaptation protocol (SDAP), or the like. Each control function can be implemented with an interface configured to communicate signals with other control functions hosted by the CU **210**. The CU **210** may be configured to handle user plane functionality (i.e., Central Unit—User Plane (CU-UP)), control plane functionality (i.e., Central Unit—Control Plane (CU-CP)), or a combination thereof. In some implementations, the CU **210** can be logically split into one or more CU-UP units and one or more CU-CP units. The CU-UP unit can communicate bidirectionally with the CU-CP unit via an interface, such as the E1 interface when implemented in an O-RAN configuration. The CU **210** can be implemented to communicate with the DU **230**, as necessary, for network control and signaling.

[0052] The DU **230** may correspond to a logical unit that includes one or more base station functions to control the operation of one or more RUs **240**. In some aspects, the DU **230** may host one or more of a radio link control (RLC) layer, a medium access control (MAC) layer, and one or more high physical (PHY) layers (such as modules for forward error correction (FEC) encoding and decoding, scrambling, modulation and demodulation, or the like) depending, at least in part, on a functional split, such as those defined by the third Generation Partnership Project (3GPP). In some aspects, the DU **230** may further host one or more low PHY layers. Each layer (or module) can be implemented with an interface configured to communicate signals with other layers (and modules) hosted by the DU **230**, or with the control functions hosted by the CU **210**.

[0053] Lower-layer functionality can be implemented by one or more RUs **240**. In some deployments, an RU **240**, controlled by a DU **230**, may correspond to a logical node that hosts RF processing functions, or low-PHY layer functions (such as performing fast Fourier transform (FFT), inverse FFT (iFFT), digital beamforming, physical random access channel (PRACH) extraction and filtering, or the like), or both, based at least in part on the functional split, such as a lower layer functional split. In such an architecture, the RU(s) **240** can be implemented to handle over the air (OTA) communication with one or more UEs **104**. In some implementations, real-time and non-real-time aspects of control and user plane communication with the RU(s) **240** can be controlled by the corresponding DU **230**. In some scenarios, this configuration can enable the DU(s) **230** and the CU **210** to be implemented in a cloud-based RAN architecture, such as a vRAN architecture.

[0054] The SMO Framework **205** may be configured to support RAN deployment and provisioning of non-virtualized and virtualized network elements. For non-virtualized network elements, the SMO Framework **205** may be configured to support the deployment of dedicated physical resources for RAN coverage requirements which may be managed via an operations and maintenance interface (such as an O1 interface). For virtualized network elements, the SMO Framework **205** may be configured to interact with a cloud computing platform (such as an open cloud (O-Cloud) **290**) to perform network element life cycle management (such as to instantiate virtualized network elements) via a cloud computing platform interface (such as an O2 interface). Such virtualized network elements can include, but are not limited to, CUs **210**, DUs **230**, RUs **240** and Near-RT RICs **225**. In some implementations, the SMO Framework **205** can communicate with a hardware aspect of a 4G RAN, such as an open eNB (O-eNB) **211**, via an O1 interface. Additionally, in some implementations, the SMO Framework **205** can communicate directly with one or more RUs **240** via an O1 interface. The SMO Framework **205** also may include a Non-RT RIC **215** configured to support functionality of the SMO Framework **205**.

[0055] The Non-RT RIC **215** may be configured to include a logical function that enables non-real-time control and optimization of RAN elements and resources, Artificial Intelligence/Machine Learning (AI/ML) workflows including model training and updates, or policy-based guidance of applications/features in the Near-RT RIC **225**. The Non-RT RIC **215** may be coupled to or communicate with (such as via an A1 interface) the Near-RT RIC **225**. The Near-RT RIC **225** may be configured to include a logical function that enables near-real-time control and optimization of RAN elements and resources via data collection and actions over an interface (such as via an E2 interface) connecting one or more CUs **210**, one or more DUs **230**, or both, as well as an O-eNB, with the Near-RT RIC **225**.

[0056] In some implementations, to generate AI/ML models to be deployed in the Near-RT RIC **225**, the Non-RT RIC **215** may receive parameters or external enrichment information from external servers. Such information may be utilized by the Near-RT RIC **225** and may be received at the SMO Framework **205** or the Non-RT RIC **215** from non-network data sources or from network functions. In some examples, the Non-RT RIC **215** or the Near-RT RIC **225** may be configured to tune RAN behavior or performance.

[0057] For example, the Non-RT RIC **215** may monitor long-term trends and patterns for performance and employ AI/ML models to perform corrective actions through the SMO Framework **205** (such as reconfiguration via **01**) or via creation of RAN management policies (such as A1 policies).

[0058] Turning now to FIGS. **3-9**, aspects are depicted with reference to one or more components and one or more methods that may perform the actions or operations described herein, where aspects in dashed line may be optional. Although the operations described below in FIGS. **7** and **8** are presented in a particular order and/or as being performed by an example component, it should be understood that the ordering of the actions and the components performing the actions may be varied, depending on the implementation. Moreover, it should be understood that the following actions, functions, and/or described components may be performed by a specially programmed processor, a processor executing specially programmed software or computer-readable media, or by any other combination of a hardware component and/or a software component capable of performing the described actions or functions.

[0059] Referring to FIG. **3**, one example of an implementation of UE **104** may include a variety of components, some of which have already been described above and are described further herein, including components such as one or more processors **312** and one or more memories **316** and one or more transceivers **302** in communication via one or more buses **344**. For example, the one or more processors **312** can include a single processor or multiple processors configured to perform one or more functions described herein. For example, the multiple processors can be configured to perform a certain subset of a set of functions described herein, such that the multiple processors together can perform the set of functions. Similarly, for example, the one or more memories **316** can include a single memory device or multiple memory devices configured to store instructions or parameters for performing one or more functions described herein. For example, the multiple memory devices can be configured to store the instructions or parameters for performing a certain subset of a set of functions described herein, such that the multiple memory devices together can store the instructions or parameters for the set of functions. The one or more processors **312**, one or more memories **316**, and one or more transceivers **302** may operate in conjunction with modem **340** and/or UE communicating component **342** for reporting available energy information to a network entity for receiving an associated scheduling decision, in accordance with aspects described herein.

[0060] In an aspect, the one or more processors **312** can include a modem **340** and/or can be part of the modem **340** that uses one or more modem processors. Thus, the various functions related to UE communicating component **342** may be included in modem **340** and/or processors **312** and, in an aspect, can be executed by a single processor, while in other aspects, different ones of the functions may be executed by a combination of two or more different processors. For example, in an aspect, the one or more processors **312** may include any one or any combination of a modem processor, or a baseband processor, or a digital signal processor, or a transmit processor, or a receiver processor, or a transceiver processor associated with transceiver **302**. In other aspects, some of the features of the one or more processors **312** and/or modem **340** associated with UE communicating component **342** may be performed by transceiver **302**.

[0061] Also, memory/memories **316** may be configured to store data used herein and/or local versions of applications **375** or UE communicating component **342** and/or one or more of its subcomponents being executed by at least one processor **312**. Memory/memories **316** can include any type of computer-readable medium usable by a computer or at least one processor **312**, such as random access memory (RAM), read only memory (ROM), electronically erasable programmable ROM (EEPROM), tapes, volatile memory, non-volatile memory, optical disk storage, magnetic disk storage, other magnetic storage devices, combinations of the aforementioned types of computer-readable media, or any other medium that can be used to store computer executable code in the form of instructions or data structures that can be accessed by a computer. In an aspect, for example, memory/memories **316** may be a non-transitory computer-readable storage medium that stores one or more computer-executable codes defining UE communicating component **342** and/or one or more of its subcomponents, and/or data associated therewith, when UE **104** is operating at least one processor **312** to execute UE communicating component **342** and/or one or more of its subcomponents.

[0062] Transceiver **302** may include at least one receiver **306** and at least one transmitter **308**. Receiver **306** may include hardware, firmware, and/or software code executable by a processor for receiving data, the code comprising instructions and being stored in a memory (e.g., computer-readable medium). Receiver **306** may be, for example, a radio frequency (RF) receiver. In an aspect, receiver **306** may receive signals transmitted by at least one base station **102**. Additionally, receiver **306** may process such received signals, and also may obtain measurements of the signals, such as, but not limited to, Ec/Io, signal-to-noise ratio (SNR), signal-to-interference-and-noise ratio (SINR), reference signal received power (RSRP), reference signal received quality (RSRQ), received signal strength indicator (RSSI), etc. Transmitter **308** may include hardware, firmware, and/or software code executable by a processor for transmitting data, the code comprising instructions and being stored in a memory (e.g., computer-readable medium). A suitable example of transmitter **308** may including, but is not limited to, an RF transmitter.

[0063] Moreover, in an aspect, UE **104** may include RF front end **388**, which may operate in communication with one or more antennas **365** and transceiver **302** for receiving and transmitting radio transmissions, for example, wireless communications transmitted by at least one base station **102** or wireless transmissions transmitted by UE **104**. RF front end **388** may be connected to one or more antennas **365** and can include one or more low-noise amplifiers (LNAs) **390**, one or more switches **392**, one or more power amplifiers (PAs) **398**, and one or more filters **396** for transmitting and receiving RF signals.

[0064] In an aspect, LNA **390** can amplify a received signal at a desired output level. In an aspect, each LNA **390** may have a specified minimum and maximum gain values. In an aspect, RF front end **388** may use one or more switches **392** to select a particular LNA **390** and its specified gain value based on a desired gain value for a particular application.

[0065] Further, for example, one or more PA(s) **398** may be used by RF front end **388** to amplify a signal for an RF output at a desired output power level. In an aspect, each PA **398** may have specified minimum and maximum gain values. In an aspect, RF front end **388** may use one or more switches **392** to select a particular PA **398** and its specified gain value based on a desired gain value for a particular application.

[0066] Also, for example, one or more filters **396** can be used by RF front end **388** to filter a received signal to obtain an input RF signal. Similarly, in an aspect, for example, a respective filter **396** can be used to filter an output from a respective PA **398** to produce an output signal for transmission. In an aspect, each filter **396** can be connected to a specific LNA **390** and/or PA **398**. In an aspect, RF front end **388** can use one or more switches **392** to select a transmit or receive path using a specified filter **396**, LNA **390**, and/or PA **398**, based on a configuration as specified by transceiver **302** and/or processor **312**.

[0067] As such, transceiver **302** may be configured to transmit and receive wireless signals through one or more antennas **365** via RF front end **388**. In an aspect, transceiver may be tuned to operate at specified frequencies such that UE **104** can communicate with, for example, one or more base stations **102** or one or more cells associated with one or more base stations **102**. In an aspect, for example, modem **340** can configure transceiver **302** to operate at a specified frequency and power level based on the UE configuration of the UE **104** and the communication protocol used by modem **340**.

[0068] In an aspect, modem **340** can be a multiband-multimode modem, which can process digital data and communicate with transceiver **302** such that the digital data is sent and received using transceiver **302**. In an aspect, modem **340** can be multiband and be configured to support multiple frequency bands for a specific communications protocol. In an aspect, modem **340** can be multimode and be configured to support multiple operating networks and communications protocols. In an aspect, modem **340** can control one or more components of UE **104** (e.g., RF front end **388**, transceiver **302**) to enable transmission and/or reception of signals from the network based on a specified modem configuration. In an aspect, the modem configuration can be based on the mode of the modem and the frequency band in use. In another aspect, the modem configuration can be based on UE configuration information associated with UE **104** as provided by the network during cell selection and/or cell reselection.

[0069] In an aspect, UE communicating component **342** can optionally include a parameter indicating component **352** for indicating one or more parameters related to an available energy at the UE **104**, a scheduling processing component **354** for receiving and/or processing a scheduling decision and/or one or more uplink grants received from a network entity, and/or a power allocating component **356** for allocating transmit power for transmitting uplink communications over resources of the uplink grants based on the scheduling decision and/or the available energy, in accordance with aspects described herein.

[0070] In an aspect, the processor(s) **312** may correspond to one or more of the processors described in connection with the UE in FIG. **9**. Similarly, the memory/memories **316** may correspond to the one or more memories described in connection with the UE in FIG. **9**.

[0071] Referring to FIG. **4**, one example of an implementation of base station **102** (e.g., a base station **102** and/or gNB **180**, as described above) may include a variety of components, some of which have already been described above, but including components such as one or more processors **412** and one or more memories **416** and one or more transceivers **402** in communication via one or more buses **444**. For example, the one or more processors **412** can include a single processor or multiple processors configured to perform one or more functions described herein. For example, the multiple processors can be configured to perform a certain subset of a set of functions described herein, such that the multiple processors together can perform the set of functions. Similarly, for example, the one or more memories **416** can include a single memory device or multiple memory devices configured to store instructions or parameters for performing one or more functions described herein. For example, the multiple memory devices can be configured to store the instructions or parameters for performing a certain subset of a set of functions described herein, such that the multiple memory devices together can store the instructions or parameters for the set of functions. The one or more processors **412**, one or more memories **416**, and one or more transceivers **402** may operate in conjunction with modem **440** and/or BS communicating component **442** for scheduling resources for a UE based on received available energy information for the UE, in accordance with aspects described herein.

[0072] The transceiver **402**, receiver **406**, transmitter **408**, one or more processors **412**, memory/memories **416**, applications **475**, buses **444**, RF front end **488**, LNAs **490**, switches **492**, filters **496**, PAs **498**, and one or more antennas **465** may be the same as or similar to the corresponding components of UE **104**, as described above, but configured or otherwise programmed for base station operations as opposed to UE operations.

[0073] In an aspect, BS communicating component **442** can optionally include a parameter processing component **452** for receiving and/or processing one or more parameters related to available energy at a UE **104**, and/or a scheduling component **454** for generating and/or transmitting, for the UE **104**, a scheduling decision and/or one or more corresponding uplink grants, in accordance with aspects described herein.

[0074] In an aspect, the processor(s) **412** may correspond to one or more of the processors described in connection with the base station in FIG. **9**. Similarly, the memory/memories **416** may correspond to the one or more memories described in connection with the base station in FIG. **9**.

[0075] FIG. **5** illustrates an example of a scheduler architecture **500** for a node scheduling a UE for wireless communications based on an indicated available energy at the UE, in accordance with aspects described herein. For example, a base station, gNB, sidelink UE, or substantially any node or network entity that schedules resources for a UE to transmit wireless communications can utilize the scheduler architecture **500**. In an example, scheduler architecture **500** can include an outer scheduler **502** and an inner scheduler **504** and/or one or more additional scheduler. For example, each scheduler **502** and **504** can be a process or loop that executes according to a time period (e.g., at a periodicity) to generate scheduling decisions for a UE. For example, the outer scheduler **502** can run once every number, T, of milliseconds (ms), and the inner scheduler **504** can run every slot (or otherwise more frequently than the outer scheduler). For example, a slot in 5G NR can include a collection of multiple symbols (e.g., orthogonal frequency division multiplexing (OFDM) symbols, single carrier-frequency division multiplexing (SC-FDM) symbols, and/or the like).

[0076] As shown in FIG. **5**, for example, the outer scheduler **502** can set long-term scheduling behavior, as described above, and can take energy availability at the UE, buffer status at the UE, etc., into account in generating scheduling decisions and/or parameters for the inner scheduler **504** to use in generating scheduling decisions. Inner scheduler **504**, for example, can make more immediate scheduling decisions (e.g., for a slot) and may take fairness among UEs or other considerations into account. In any case, inner scheduler **504** can generate per slot scheduling decisions for a UE, which can be based on one or more parameters from the outer scheduler **502**. For example, the outer scheduler **502** can manage scheduling metrics that are influenced by more slowly varying input from UEs such as energy availability, buffer status, latency requirements, etc., as described further herein. In an example, the outer scheduler **502** can set overall scheduling policy guidelines for the inner scheduler **504** for a given period of time. In an example, the outer scheduler **502** can influence slot-level decisions made by the inner scheduler **504**. For example, the outer scheduler **502** may cap the maximum duty cycle at which a UE can be engaged in uplink. This cap may prohibit the inner scheduler **504** from scheduling that UE too often.

[0077] FIG. **6** illustrates an example of an outer scheduler **600** architecture in a gNB, in accordance with aspects described herein. For example, the outer scheduler **600** architecture can be used by outer scheduler **502** or substantially any node scheduling a UE for wireless communications, as described herein. In an example, the gNB outer scheduler **600** can be configured with certain parameters at **602**, such as a period (T) in ms and a number of slots (K) per T, a buffer size threshold (V) to allow even power distribution at a UE, etc. For example, a UE can inform a gNB of a buffer size and available energy. The gNB outer scheduler **600** can receive the indication of the buffer size and available energy of the UE and can compare the buffer size (S) to a threshold (V) at **604**.

[0078] For example, if S>V, at **606**, the gNB outer scheduler **600** can set the target duty cycle=1 and the UE transmission (Tx) power=E/K, where E is the available energy reported by the UE. For example, if S<=V, at **608**, the gNB can estimate an optimal duty cycle to dequeue the uplink buffer using a fewest number of uplink grants, set P.sub.0=E/L, where L is a smallest number of slots such that L*B>S, and can set B (bits/second)=W*log (1+SINR) where W is the UE allocated bandwidth, and SINR is the SINR of a signal received from the UE. In either case, at **610**, the gNB can set the target duty cycle for the next time period and can convey the assigned UE Tx power to the UE for use in transmitting uplink communications over a sequence of one or more uplink grants provided to the UE by the gNB.
[0079] For example, where the goal is to dequeue the buffer as efficiently as possible, two variables can be considered—L: dictates duty cycle (e.g., number of scheduled slots in a certain period of time), and W: dictates frequency domain allocation. A goal can be to minimize L subject to the following:
[00001]$P = \frac{E}{L}$ .fwdarw. transmitpowerconstraint$\frac{E}{L} \leq \text{Pmax}$ .fwdarw. maxtransmitpowerlimit [0080] L*Ts*B≥S'buffer requirement, where Ts is slot duration [0081] W<W.sub.max.fwdarw.bandwidth constraint
[00002]$B = W * \log(1 + \frac{P}{N * W})$ .fwdarw. linkcapacity,
where N is noise density (e.g., milliwatt/Hz)
[00003]$\frac{P}{N * W} > \text{SNR}_{\text{min}}$ .fwdarw. minimumlinkqualityrequirement [0082] L<K.fwdarw.duty cycle bound
[0083] For example, E, N,SNR.sub.min can be constants while the variables can include L, P, B, and W.
[0084] In this example, using the above terms, the outer scheduler **502** or **600** can select a buffer size threshold (V) as follows (e.g., when buffer size is large, it may be best to use all slots): [0085] Set L=K [0086] Then,
[00004]$P_{opt} = \min(\text{Pmax}, \frac{E}{K})B = W * \log(1 + \frac{P_{\text{opt}}}{N * W})\frac{P_{\text{opt}}}{N * W} > \text{SNR}_{\text{min}}$
$K * \text{Ts} * B = K * \text{Ts} * W * \log(1 + \frac{P_{\text{opt}}}{N * W})\text{Set}W = W_0 = \min(W_{\text{max}}, \frac{P_{\text{opt}}}{N * \text{SNRmin}})$
So$K * \text{Ts} * B = K * Ts * W0 * \log(1 + \frac{P_{\text{opt}}}{N * W0})$
Thus, setting
[00005]$V = K * \text{Ts} * W0 * \log(1 + \frac{P_{\text{opt}}}{N * W0})$
can yield the criterion to set target duty cycle to be 1 (e.g., where S>V).
[0087] In addition, for example, using the above terms, the outer scheduler **502** or **600** can select the target duty cycle when S<=V as follows: [0088] Minimize: L [0089] Subject to:
[00006]$L * \text{Ts} * W * \log(1 + \frac{E}{L * N * W}) \geq S \frac{E}{L * N * W} > \text{SNR}_{\text{min}}$ [0090] W≤W.sub.max (where W.sub.max is the total available BW)
[00007]$\frac{E}{L} \leq \text{Pmax}$ [0091] Variables: L, W
[00008]Set$P_{\text{eff}} = \min(\frac{E}{L}, \text{Pmax})SetW_{\text{opt}} = \min(\frac{P_{\text{eff}}}{N * \text{SNRmin}}, W_{\text{max}})$ [0092] Constants: E, N,SNR.sub.min
[00009]$L * \text{Ts} * W_{\text{opt}} * \log(1 + \frac{P_{\text{eff}}}{N * W_{\text{opt}}}) \geq S$ [0093] It may be possible to reduce the problem to optimize over a single variable L. A binary search for optimal L may suffice to identify the optimal value of L to use. The left hand size of the last criterion, L*Ts*W_opt*log(1+P_eff/(N*W_opt)) monotonically increases in L. This behavior may have two distinct parts—a region where it monotonically increases in L followed by a region where it turns out to be a constant. In an example, the outer scheduler **502** or **600** can find the smallest duty cycle that satisfies all the constraints. Lopt/K can give the desired/target duty cycle for the medium term.
[0094] FIG. **7** illustrates a flow chart of an example of a method **700** for reporting available energy information for receiving a scheduling decision, in accordance with aspects described herein. FIG. **8** illustrates a flow chart of an example of a method **800** for scheduling a UE based on receiving available energy information regarding the UE, in accordance with aspects described herein. In an

example, a UE **104** can perform the functions described in method **700** shown in FIG. **7** using one or more of the components described in FIGS. **1** and/or **3**. In an example, a node scheduling the UE **104** with communication resources, such as a base station **102** or gNB **180**, a monolithic base station or gNB, a portion of a disaggregated base station or gNB, a UE in sidelink communication, etc., can perform the functions described in method **800** shown in FIG. **8** using one or more of the components described in FIGS. **1** and/or **4**. Methods **700** and **800** are described in conjunction with one another for ease of explanation; however, the methods **700** and **800** are not required to be performed together and indeed can be performed independently using separate devices.

[0095] In method **700**, at Block **702**, an indication of available energy for a period of time and a buffer size can be transmitted for a network entity. In an aspect, parameter indicating component **352**, e.g., in conjunction with processor(s) **312**, memory/memories **316**, transceiver **302**, UE communicating component **342**, etc., can transmit, for a network entity, an indication of available energy for a period of time and a buffer size. For example, parameter indicating component **352** can transmit, to the network entity, an indication of the available energy (E) and the buffer size (S) in uplink signaling, such as in uplink control information (UCI) transmitted by the UE **104** to the network entity over a physical uplink control channel (PUCCH), physical uplink shared channel (PUSCH), etc. For example, the indication of available energy can include, or may be computed based on, a power headroom for one or more future time periods based on a time averaging window including the one or more future time periods and multiple previous time periods, as described above.

[0096] In method **700**, optionally at Block **704**, the available energy can be estimated based on a history of transmit power used in multiple previous periods of time. In an aspect, parameter indicating component **352**, e.g., in conjunction with processor(s) **312**, memory/memories **316**, transceiver **302**, UE communicating component **342**, etc., can estimate the available energy based on the history of transmit power used in multiple previous periods of time. As described above, for example, parameter indicating component **352** can estimate the available energy for the one or more future time periods (e.g., for which available energy is reported) by determining the available energy that would comply with a SAR (or PD) limit over the one or more future time periods and multiple previous time periods in a time averaging window. In one example, the time periods can correspond to slots, as defined in 5G NR.

[0097] In method **800**, at Block **802**, an indication of available energy for a period of time and a buffer size can be received for a UE. In an aspect, parameter processing component **452**, e.g., in conjunction with processor(s) **412**, memory/memories **416**, transceiver **402**, BS communicating component **442**, etc., can receive and/or process, for the UE, the indication of available energy for the period of time and the buffer size. For example, parameter processing component **452** can receive the indication of available energy in UCI transmitted by the UE **104** (e.g., in a PUCCH, PUSCH, etc.), which can correspond to other resources granted for the UE **104** by the network entity. The indication of available energy can be in terms of millijoules (mJ), or substantially any other measure of energy.

[0098] In method **800**, at Block **804**, a scheduling decision including a target duty cycle for the UE can be generated, for the UE, based on the available energy and the buffer size at the UE. In an aspect, scheduling component **454**, e.g., in conjunction with processor(s) **412**, memory/memories **416**, transceiver **402**, BS communicating component **442**, etc., can generate, for the UE and based on the available energy and the buffer size at the UE, the scheduling decision including the target duty cycle for the UE. For example, scheduling component **454** can generate the scheduling decision based on the outer scheduling process (e.g., outer scheduler **502** or **600**) that sets one or more long term scheduling parameters based on the available energy and the buffer status and an inner scheduling process (e.g., inner scheduler **504**) that executes more frequently than the outer scheduling process and generates the scheduling decision based on the one or more long term scheduling parameters and one or more other parameters. For example, scheduling component **454**

can execute the outer scheduler as a process or loop less frequently than executing the inner scheduler as a process or loop, and the outer scheduler can provide parameters to the inner scheduler as input for the inner scheduler to determine slot-level (or other time instance-level) scheduling decisions.
[0099] In one example, as described above, scheduling component **454** can use the outer scheduler to generate a long-term scheduling decision for the UE **104** based on the available energy and/or buffer size received for the UE (e.g., at Block **802**). For example, as described above, scheduling component **454** can generate the long-term scheduling decision as the target duty cycle (e.g., L) for the next one or more time periods (e.g., number of slots K). In an example, scheduling component **454** can use the inner scheduler to generate a short-term scheduling decision (e.g., per slot) for scheduling the UE **104**, which can be based on the target duty cycle computed for the UE by the outer scheduler and one or more other parameters, such as ensuring fairness in resource scheduling for multiple UEs. The scheduling decision generated by the scheduling component **454** may include one or more parameters for determining the transmit power allocation for the UE **104**.
[0100] In method **800**, optionally at Block **806**, a second indication related to the scheduling decision, based on the available energy and the buffer size at the UE, can be transmitted for the UE. In an aspect, scheduling component **454**, e.g., in conjunction with processor(s) **412**, memory/memories **416**, transceiver **402**, BS communicating component **442**, etc., can transmit, for the UE and based on the available energy and the buffer size at the UE, the second indication related to the scheduling decision. For example, the second indication related to the scheduling decision can include a target duty cycle for the UE, an indication of transmit power, and/or other parameters the UE can use to determine how to allocate transmit power. For example, the second indication related to the scheduling decision may include a value indicating to allocate the transmit power as an energy level divided by a number of slots in the period of time, a value indicating a scaling factor, greater than one, for multiplying the energy level to allocate the transmit power, a value indicating a scaling factor divided by a number of slots in the period of time for multiplying the energy level to allocate the transmit power in each of the number of slots, a value indicating to allocate the transmit power as a maximum transmit power, a value indicating to limit a maximum transmit power for the transmit power at least a portion of slots in the period of time, a value indicating a target duty cycle as computed by the network entity for the UE over the period of time, etc.
[0101] In an example, scheduling component **454** can transmit the second indication related to the scheduling decision to the UE **104** to indicate one or more parameters of the outer scheduler or inner scheduler, such as target duty cycle, L, or parameters related thereto, so the UE can fine tune its strategy on how to allocate the available energy to individual scheduled slots. In one example, scheduling component **454** can transmit the second indication related to the scheduling decision to the UE using media access control (MAC)-control element (CE) signaling, downlink control information (DCI), etc. In one example, where the scheduling decision is not grant-specific, MAC-CE may be used instead of DCI. In a specific example, scheduling component **454** can use three bits to convey the second indication related to the scheduling decision to the UE **104** (e.g., an indication of an optimal way to use the available energy (E) at the UE for the next time period (T)). In this example, scheduling component **454** may use the following bit values:
TABLE-US-00001 3-bits Message Instructions 000 Allocate the UE Tx power as E/K 001-110 Allocate the UE Tx power as (E * A.sub.i) / K, where A.sub.i is s scaling factor (greater than 1) associated with the ith index slot, where the 3-bit value can map to a value for A.sub.i 111 Use P.sub.max until the UE runs out of energy
[0102] In method **700**, optionally at Block **706**, a second indication related to a scheduling decision that is based on the available energy and/or the buffer size can be received from the network entity based on transmitting the indication. In an aspect, scheduling processing component **354**, e.g., in conjunction with processor(s) **312**, memory/memories **316**, transceiver **302**, UE communicating

component **342**, etc., can receive, from the network entity and based on transmitting the indication of available energy, the second indication related to the scheduling decision that is based on the available energy and/or the buffer size. For example, the second indication can include a target duty cycle or an indication on how to allocate the transmit power at the UE **104**, in accordance with various examples described above. In one specific example, the second indication can include target duty cycle, L, or parameters related thereto, so the UE can fine tune its strategy on how to allocate the available energy to individual scheduled slots. In one example described above, the second indication may include the three-bit indicator, where the value indicates an instruction for allocating transmit power.

[0103] In method **800**, at Block **808**, a sequence of one or more scheduling grants indicating resources for the UE to use in transmitting the communications in the period of time can be transmitted for the UE and in accordance with the scheduling decision. In an aspect, scheduling component **454**, e.g., in conjunction with processor(s) **412**, memory/memories **416**, transceiver **402**, BS communicating component **442**, etc., can transmit, for the UE and in accordance with the scheduling decision, the sequence of one or more scheduling grants indicating resources for the UE to use in transmitting the communications in the period of time. For example, scheduling component **454** can transmit the sequence of one or more scheduling grants to the UE **104** in DCI (e.g., over physical downlink control channel (PDCCH) or physical downlink shared channel (PDSCH) resources). The sequence of one or more scheduling grants can include resources that comply with the scheduling decision (e.g., the target duty cycle computed for the UE **104**) over the period of time (e.g., one or more future periods of time) to allow the UE **104** to use resources and comply with SAR (or PD) limits.

[0104] In method **700**, at Block **708**, a sequence of one or more scheduling grants indicating resources for the UE to use in transmitting the communications in the period of time can be received from the network entity and based on transmitting the indication of the available energy. In an aspect, scheduling processing component **354**, e.g., in conjunction with processor(s) **312**, memory/memories **316**, transceiver **302**, UE communicating component **342**, etc., can receive, from the network entity and based on the transmitting the indication of the available energy, the sequence of one or more scheduling grants indicating resources for the UE to use in transmitting the communications in the period of time. For example, scheduling processing component **354** can receive the sequence of one or more uplink grants in DCI from the network entity, and the uplink grants may include resources that are allocated based on the available energy and/or buffer size indicated by the UE.

[0105] In method **700**, at Block **710**, the transmit power for transmitting the communications in the resources over the period of time or one or more additional periods of time can be allocated based on the available energy and/or the second indication, and according to the sequence of one or more scheduling grants. In an aspect, power allocating component **356**, e.g., in conjunction with processor(s) **312**, memory/memories **316**, transceiver **302**, UE communicating component **342**, etc., can allocate, based on the available energy and/or the second indication, and according to the sequence of one or more scheduling grants, the transmit power for transmitting the communications in the resources over the period of time or one or more additional periods of time. For example, as described, power allocating component **356** can allocate the transmit power over the period of time or one or more additional periods of time so the average transmit power over the period of time or one or more additional periods (and/or previous periods of time) complies with a SAR (or PD) limit. Moreover, as described, this can include power allocating component **356** boosting power in some time periods.

[0106] In one example, where the scheduling processing component **354** receives the second indication related to the scheduling decision, power allocating component **356** can also use the second indication in allocating the transmit power. For example, power allocating component **356** can use an indicated target duty cycle to allocate the transmit power. In another example, power allocating component **356** can use the instructions indicated by a value in the three-bit message to allocate the transmit power. In other examples, power allocating component **356** can allocate the transmit power based on the second indication including one or more of a value indicating to allocate the transmit power as an the reported available energy divided by a number of slots in the period of time, a value indicating the scaling factor, greater than one, for multiplying the reported available energy to allocate the transmit power, a value indicating a scaling factor divided by a number of slots in the period of time for multiplying the reported available energy to allocate the transmit power in each of the number of slots, a value indicating to allocate the transmit power as a maximum transmit power, a value indicating to limit a maximum transmit power for the transmit power at least a portion of slots in the period of time, a value indicating a target duty cycle for the UE over the period of time, etc. In any case, for example, power allocating component **356** can allocate transmit power for the period of time by adjusting a power amplifier or other RF component during the duration of the period of time, to allow UE communicating component **342** to transmit uplink communications using the allocated power.

[0107] In method **700**, optionally at Block **712**, the communications can be transmitted, using the transmit power, in the period of time or one or more additional periods of time. In an aspect, UE communicating component **342**, e.g., in conjunction with processor(s) **312**, memory/memories **316**, transceiver **302**, etc., can transmit, using the transmit power, the communications in the period of time or one or more additional periods of time. For example, based on the power allocated by power allocating component **356** in a period of time (e.g., slot), UE communicating component **342** can transmit the uplink communications in the period of time.

[0108] In method **800**, optionally at Block **810**, an indication of a scaling factor can be transmitted for the UE. In an aspect, scheduling component **454**, e.g., in conjunction with processor(s) **412**, memory/memories **416**, transceiver **402**, BS communicating component **442**, etc., can transmit, for the UE, an indication of a scaling factor. For example, scheduling component **454** can transmit an indication of the scaling factor A.sub.i for the UE **104** in DCI, MAC-CE, RRC, or other signaling. In one specific example, scheduling component **454** can transmit, to the UE **104** in RRC signaling, a mapping of scaling factor values for A.sub.i to three-bit message values, as described above.

[0109] In method **700**, at Block **714**, an indication of a scaling factor can be received from the network entity. In an aspect, scheduling processing component **354**, e.g., in conjunction with processor(s) **312**, memory/memories **316**, transceiver **302**, UE communicating component **342**, etc., can receive, from the network entity, the indication of the scaling factor. As described, scheduling processing component **354** can receive the indication in DCI, MAC-CE, RRC, or other signaling, and the indication may include an indication of the scaling factor A.sub.i, or a mapping of values for the scaling factor A.sub.i to three-bit message values, etc. In one example, where the second indication received from the network entity (e.g., at Block **706**) includes using the scaling factor, scheduling processing component **354** can use the indicated scaling factor received from the network entity and/or can determine the scaling factor based on the received mapping and a value indicated by the second indication.

[0110] FIG. **9** is a block diagram of a MIMO communication system **900** including a base station **102** and a UE **104**. The MIMO communication system **900** may illustrate aspects of the wireless communication access network **100** described with reference to FIG. **1**. The base station **102** may be an example of aspects of the base station **102** described with reference to FIG. **1**. The base station **102** may be equipped with antennas **934** and **935**, and the UE **104** may be equipped with antennas **952** and **953**. In the MIMO communication system **900**, the base station **102** may be able to send data over multiple communication links at the same time. Each communication link may be called a “layer” and the “rank” of the communication link may indicate the number of layers used for communication. For example, in a 2×2 MIMO communication system where base station **102** transmits two “layers,” the rank of the communication link between the base station **102** and the UE **104** is two.

[0111] At the base station **102**, a transmit (Tx) processor **920** may receive data from a data source. The transmit processor **920** may process the data. The transmit processor **920** may also generate control symbols or reference symbols. A transmit MIMO processor **930** may perform spatial processing (e.g., precoding) on data symbols, control symbols, or reference symbols, if applicable, and may provide output symbol streams to the transmit modulator/demodulators **932** and **933**. Each modulator/demodulator **932** through **933** may process a respective output symbol stream (e.g., for OFDM, etc.) to obtain an output sample stream. Each modulator/demodulator **932** through **933** may further process (e.g., convert to analog, amplify, filter, and upconvert) the output sample stream to obtain a DL signal. In one example, DL signals from modulator/demodulators **932** and **933** may be transmitted via the antennas **934** and **935**, respectively.
[0112] The UE **104** may be an example of aspects of the UEs **104** described with reference to FIGS. **1** and **3**. At the UE **104**, the UE antennas **952** and **953** may receive the DL signals from the base station **102** and may provide the received signals to the modulator/demodulators **954** and **955**, respectively. Each modulator/demodulator **954** through **955** may condition (e.g., filter, amplify, downconvert, and digitize) a respective received signal to obtain input samples. Each modulator/demodulator **954** through **955** may further process the input samples (e.g., for OFDM, etc.) to obtain received symbols. A MIMO detector **956** may obtain received symbols from the modulator/demodulators **954** and **955**, perform MIMO detection on the received symbols, if applicable, and provide detected symbols. A receive (Rx) processor **958** may process (e.g., demodulate, deinterleave, and decode) the detected symbols, providing decoded data for the UE **104** to a data output, and provide decoded control information to a processor(s) **980**, or memory/memories **982**.
[0113] The processor(s) **980** may in some cases execute stored instructions to instantiate a UE communicating component **342** (see e.g., FIGS. **1** and **3**).
[0114] On the uplink (UL), at the UE **104**, a transmit processor **964** may receive and process data from a data source. The transmit processor **964** may also generate reference symbols for a reference signal. The symbols from the transmit processor **964** may be precoded by a transmit MIMO processor **966** if applicable, further processed by the modulator/demodulators **954** and **955** (e.g., for single carrier-FDMA, etc.), and be transmitted to the base station **102** in accordance with the communication parameters received from the base station **102**. At the base station **102**, the UL signals from the UE **104** may be received by the antennas **934** and **935**, processed by the modulator/demodulators **932** and **933**, detected by a MIMO detector **936** if applicable, and further processed by a receive processor **938**. The receive processor **938** may provide decoded data to a data output and to the processor(s) **940** or memory/memories **942**.
[0115] The processor(s) **940** may in some cases execute stored instructions to instantiate a BS communicating component **442** (see e.g., FIGS. **1** and **4**).
[0116] The components of the UE **104** may, individually or collectively, be implemented with one or more ASICs adapted to perform some or all of the applicable functions in hardware. Each of the noted modules may be a means for performing one or more functions related to operation of the MIMO communication system **900**. Similarly, the components of the base station **102** may, individually or collectively, be implemented with one or more application specific integrated circuits (ASICs) adapted to perform some or all of the applicable functions in hardware. Each of the noted components may be a means for performing one or more functions related to operation of the MIMO communication system **900**.
[0117] The following aspects are illustrative only and aspects thereof may be combined with aspects of other embodiments or teaching described herein, without limitation.
[0118] Aspect 1 is a method for wireless communication at a UE including transmitting, for a network entity, an indication of available energy at the UE for a period of time and a buffer size at the UE, receiving, from the network entity and based on transmitting the indication of the available energy, a sequence of one or more scheduling grants indicating resources for the UE to use in

transmitting the communications in the period of time, and allocating, based on the available energy and according to the sequence of one or more scheduling grants, a transmit power for transmitting the communications in the resources over the period of time or one or more additional periods of time.

[0119] In Aspect 2, the method of Aspect 1 includes receiving, from the network entity and based on transmitting the indication, a second indication related to using the available energy to allocate transmit power for transmitting communications in the period of time, where allocating the transmit power is further based on the second indication.

[0120] In Aspect 3, the method of Aspect 2 includes where the second indication includes a value indicating to allocate the transmit power as an energy level divided by a number of slots in the period of time.

[0121] In Aspect 4, the method of any of Aspects 2 or 3 includes where the second indication includes a value indicating a scaling factor, greater than one, for multiplying the available energy to allocate the transmit power.

[0122] In Aspect 5, the method of Aspect 4 includes receiving, from the network entity, a third indication of the scaling factor in RRC signaling.

[0123] In Aspect 6, the method of any of Aspects 2 to 5 includes where the second indication includes a value indicating a scaling factor divided by a number of slots in the period of time for multiplying the available energy to allocate the transmit power in each of the number of slots.

[0124] In Aspect 7, the method of any of Aspects 2 to 6 includes where the second indication includes a value indicating to allocate the transmit power as a maximum transmit power.

[0125] In Aspect 8, the method of any of Aspects 2 to 7 includes where the second indication includes a value indicating to limit a maximum transmit power for the transmit power at least a portion of slots in the period of time.

[0126] In Aspect 9, the method of any of Aspects 2 to 8 includes where the second indication includes a value indicating a target duty cycle for the UE over the period of time.

[0127] In Aspect 10, the method of any of Aspects 2 to 9 includes where receiving the second indication includes receiving a MAC-CE including the second indication.

[0128] In Aspect 11, the method of any of Aspects 1 to 10 includes estimating the available energy based on a history of transmit power used by the UE in multiple previous periods of time.

[0129] Aspect 12 is a method for wireless communication at a network node including receiving, for a UE, an indication of available energy at the UE for a period of time and a buffer size at the UE, generating, for the UE and based on the available energy and the buffer size at the UE, a scheduling decision including a target duty cycle for the UE, and transmitting, for the UE and in accordance with the scheduling decision, a sequence of one or more scheduling grants indicating resources for the UE to use in transmitting the communications in the period of time.

[0130] In Aspect 13, the method of Aspect 12 includes transmitting, for the UE and based on the available energy and the buffer size at the UE, a second indication related to the scheduling decision.

[0131] In Aspect 14, the method of Aspect 13 includes where the second indication includes a value indicating to allocate a transmit power as the available energy divided by a number of slots in the period of time.

[0132] In Aspect 15, the method of any of Aspects 13 or 14 includes where the second indication includes a value indicating a scaling factor, greater than one, for multiplying the available energy to allocate a transmit power.

[0133] In Aspect 16, the method of Aspect 15 includes transmitting, for the UE, a third indication of the scaling factor in RRC signaling.

[0134] In Aspect 17, the method of any of Aspects 13 to 16 includes where the second indication includes a value indicating a scaling factor divided by a number of slots in the period of time for multiplying the available energy to allocate a transmit power in each of the number of slots.

[0135] In Aspect 18, the method of any of Aspects 13 to 17 includes where the second indication includes a value indicating to allocate a transmit power as a maximum transmit power.
[0136] In Aspect 19, the method of any of Aspects 13 to 18 includes where the second indication includes a value indicating to limit a maximum transmit power for a transmit power at least a portion of slots in the period of time.
[0137] In Aspect 20, the method of any of Aspects 13 to 19 includes where the second indication includes a value indicating a target duty cycle of the UE over the period of time.
[0138] In Aspect 21, the method of any of Aspects 13 to 20 includes where transmitting the includes transmitting a MAC-CE including the second indication.
[0139] In Aspect 22, the method of any of Aspects 12 to 21 includes where generating the scheduling decision is based on an outer scheduling process that sets one or more long term scheduling parameters based on the available energy and the buffer size, and an inner scheduling process that executes more frequently than the outer scheduling process and generates the scheduling decision based on the one or more long term scheduling parameters and one or more other parameters.
[0140] In Aspect 23, the method of Aspect 22 includes where the one or more long term scheduling parameters include a maximum duty cycle at which the UE can transmit the communications in the period of time.
[0141] In Aspect 24, the method of any of Aspects 22 or 23 includes where the outer scheduling process sets the one or more long term scheduling parameters based on comparing the buffer size to a threshold.
[0142] In Aspect 25, the method of Aspect 24 includes where the outer scheduling process sets the one or more long term scheduling parameters including a target duty cycle to a value of 1 and a UE transmit power for the UE to a value of the available energy divided by a number of slots in the period of time when the buffer size is larger than the threshold.
[0143] In Aspect 26, the method of any of Aspects 24 or 25 includes where the outer scheduling process sets the one or more long term scheduling parameters including a target duty cycle to a smallest value that satisfies one or more constraints associated with the available energy and the buffer size when the buffer size is smaller than the threshold.
[0144] In Aspect 27, the method of any of Aspects 24 to 26 includes where the threshold is based on the available energy, the buffer size, a number of slots in the period of time, a maximum transmit power capability, and a bandwidth constraint.
[0145] Aspect 28 is an apparatus for wireless communication including one or more processors, one or more memories coupled with the one or more processors, and instructions stored in the one or more memories and operable, when executed by the one or more processors, to cause the apparatus to perform any of the methods of Aspects 1 to 27.
[0146] Aspect 29 is an apparatus for wireless communication including means for performing any of the methods of Aspects 1 to 27.
[0147] Aspect 30 is one or more computer-readable media including code executable by one or more processors for wireless communications, the code including code for performing any of the methods of Aspects 1 to 27.
[0148] The above detailed description set forth above in connection with the appended drawings describes examples and does not represent the only examples that may be implemented or that are within the scope of the claims. The term “example,” when used in this description, means “serving as an example, instance, or illustration,” and not “preferred” or “advantageous over other examples.” The detailed description includes specific details for the purpose of providing an understanding of the described techniques. These techniques, however, may be practiced without these specific details. In some instances, well-known structures and apparatuses are shown in block diagram form in order to avoid obscuring the concepts of the described examples.
[0149] Information and signals may be represented using any of a variety of different technologies

and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, computer-executable code or instructions stored on a computer-readable medium, or any combination thereof.

[0150] The various illustrative blocks and components described in connection with the disclosure herein may be implemented or performed with a specially programmed device, such as but not limited to a processor, a digital signal processor (DSP), an ASIC, a field programmable gate array (FPGA) or other programmable logic device, a discrete gate or transistor logic, a discrete hardware component, or any combination thereof designed to perform the functions described herein. A specially programmed processor may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A specially programmed processor may also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration.

[0151] The functions described herein may be implemented in hardware, software executed by a processor, firmware, or any combination thereof. If implemented in software executed by a processor, the functions may be stored on or transmitted over as one or more instructions or code on a non-transitory computer-readable medium. Other examples and implementations are within the scope and spirit of the disclosure and appended claims. For example, due to the nature of software, functions described above can be implemented using software executed by a specially programmed processor, hardware, firmware, hardwiring, or combinations of any of these. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations. Also, as used herein, including in the claims, “or” as used in a list of items prefaced by “at least one of” indicates a disjunctive list such that, for example, a list of “at least one of A, B, or C” means A or B or C or AB or AC or BC or ABC (i.e., A and B and C).

[0152] Computer-readable media includes both computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another. A storage medium may be any available medium that can be accessed by a general purpose or special purpose computer. By way of example, and not limitation, computer-readable media can comprise RAM, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium that can be used to carry or store desired program code means in the form of instructions or data structures and that can be accessed by a general-purpose or special-purpose computer, or a general-purpose or special-purpose processor. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of medium. Disk and disc, as used herein, include compact disc (CD), laser disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray disc where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above are also included within the scope of computer-readable media.

[0153] The previous description of the disclosure is provided to enable a person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be readily apparent to those skilled in the art, and the common principles defined herein may be applied to other variations without departing from the spirit or scope of the disclosure. Furthermore, although elements of the described aspects and/or embodiments may be described or claimed in the singular, the plural is contemplated unless limitation to the singular is explicitly stated. Additionally, all or a portion of any aspect and/or embodiment may be utilized with all or a portion of any other aspect and/or embodiment, unless stated otherwise. Thus, the disclosure is not to be limited to the examples and designs described herein but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

Claims

1. An apparatus for wireless communication, comprising: a transceiver; one or more memories configured to, individually or in combination, store instructions; and one or more processors communicatively coupled with the one or more memories, wherein the one or more processors are, individually or in combination, configured to execute the instructions to cause the apparatus to: transmit, for a network entity, an indication of available energy at the apparatus for a period of time and a buffer size at the apparatus; receive, from the network entity and based on transmitting the indication of the available energy, a sequence of one or more scheduling grants indicating resources for the apparatus to use in transmitting the communications in the period of time; and allocate, based on the available energy and according to the sequence of one or more scheduling grants, a transmit power for transmitting the communications in the resources over the period of time or one or more additional periods of time.
2. The apparatus of claim 1, wherein the one or more processors are, individually or in combination, configured to execute the instructions to cause the apparatus to receive, from the network entity and based on transmitting the indication, a second indication related to using the available energy to allocate transmit power for transmitting communications in the period of time, wherein allocating the transmit power is further based on the second indication.
3. The apparatus of claim 2, wherein the second indication includes a value indicating to allocate the transmit power as an energy level divided by a number of slots in the period of time.
4. The apparatus of claim 2, wherein the second indication includes a value indicating a scaling factor, greater than one, for multiplying the available energy to allocate the transmit power.
5. The apparatus of claim 4, wherein the one or more processors are, individually or in combination, configured to execute the instructions to cause the apparatus to receive, from the network entity, a third indication of the scaling factor in RRC signaling.
6. The apparatus of claim 2, wherein the second indication includes a value indicating a scaling factor divided by a number of slots in the period of time for multiplying the available energy to allocate the transmit power in each of the number of slots.
7. The apparatus of claim 2, wherein the second indication includes a value indicating to allocate the transmit power as a maximum transmit power.
8. The apparatus of claim 2, wherein the second indication includes a value indicating to limit a maximum transmit power for the transmit power at least a portion of slots in the period of time.
9. The apparatus of claim 2, wherein the second indication includes a value indicating a target duty cycle for the UE over the period of time.
10. The apparatus of claim 2, wherein the one or more processors are, individually or in combination, configured to execute the instructions to cause the apparatus to receive the second indication in a media access control (MAC)-control element (CE).
11. The apparatus of claim 1, wherein the one or more processors are, individually or in combination, configured to execute the instructions to cause the apparatus to estimate the available energy based on a history of transmit power used by the wherein the one or more processors are, individually or in combination, configured to execute the instructions to cause the apparatus to in multiple previous periods of time.
12. An apparatus for wireless communication, comprising: a transceiver; one or more memories configured to, individually or in combination, store instructions; and one or more processors communicatively coupled with the one or more memories, wherein the one or more processors are, individually or in combination, configured to execute the instructions to cause the apparatus to:

receive, for a user equipment (UE), an indication of available energy at the UE for a period of time and a buffer size at the UE; generate, for the UE and based on the available energy and the buffer size at the UE, a scheduling decision including a target duty cycle for the UE; and transmit, for the UE and in accordance with the scheduling decision, a sequence of one or more scheduling grants indicating resources for the UE to use in transmitting the communications in the period of time.
13. The apparatus of claim 12, wherein the one or more processors are, individually or in combination, configured to execute the instructions to cause the apparatus to transmit, for the UE and based on the available energy and the buffer size at the UE, a second indication related to the scheduling decision.
14. The apparatus of claim 13, wherein the second indication includes a value indicating to allocate a transmit power as the available energy divided by a number of slots in the period of time.
15. The apparatus of claim 13, wherein the second indication includes a value indicating a scaling factor, greater than one, for multiplying the available energy to allocate a transmit power.
16. The apparatus of claim 15, wherein the one or more processors are, individually or in combination, configured to execute the instructions to cause the apparatus to transmit, for the UE, a third indication of the scaling factor in radio resource control (RRC) signaling.
17. The apparatus of claim 13, wherein the second indication includes a value indicating a scaling factor divided by a number of slots in the period of time for multiplying the available energy to allocate a transmit power in each of the number of slots.
18. The apparatus of claim 13, wherein the second indication includes a value indicating to allocate a transmit power as a maximum transmit power.
19. The apparatus of claim 13, wherein the second indication includes a value indicating to limit a maximum transmit power for a transmit power at least a portion of slots in the period of time.
20. The apparatus of claim 13, wherein the second indication includes a value indicating a target duty cycle of the UE over the period of time.
21. The apparatus of claim 13, wherein the one or more processors are, individually or in combination, configured to execute the instructions to cause the apparatus to transmit the second indication in a media access control (MAC)-control element (CE).
22. The apparatus of claim 12, wherein the one or more processors are, individually or in combination, configured to execute the instructions to cause the apparatus to generate the scheduling decision based on an outer scheduling process that sets one or more long term scheduling parameters based on the available energy and the buffer size, and an inner scheduling process that executes more frequently than the outer scheduling process and generates the scheduling decision based on the one or more long term scheduling parameters and one or more other parameters.
23. The apparatus of claim 22, wherein the one or more long term scheduling parameters include a maximum duty cycle at which the UE can transmit the communications in the period of time.
24. The apparatus of claim 22, wherein the outer scheduling process sets the one or more long term scheduling parameters based on comparing the buffer size to a threshold.
25. The apparatus of claim 24, wherein the outer scheduling process sets the one or more long term scheduling parameters including a target duty cycle to a value of 1 and a UE transmit power for the UE to a value of the available energy divided by a number of slots in the period of time when the buffer size is larger than the threshold.
26. The apparatus of claim 24, wherein the outer scheduling process sets the one or more long term scheduling parameters including a target duty cycle to a smallest value that satisfies one or more constraints associated with the available energy and the buffer size when the buffer size is smaller than the threshold.
27. The apparatus of claim 24, wherein the threshold is based on the available energy, the buffer size, a number of slots in the period of time, a maximum transmit power capability, and a bandwidth constraint.

28. A method for wireless communication at a user equipment (UE), comprising: transmitting, for a network entity, an indication of available energy at the UE for a period of time and a buffer size at the UE; receiving, from the network entity and based on transmitting the indication of the available energy, a sequence of one or more scheduling grants indicating resources for the UE to use in transmitting the communications in the period of time; and allocating, based on the available energy and according to the sequence of one or more scheduling grants, a transmit power for transmitting the communications in the resources over the period of time or one or more additional periods of time.

29. The method of claim 28, further comprising receiving, from the network entity and based on transmitting the indication, a second indication related to using the available energy to allocate transmit power for transmitting communications in the period of time, wherein allocating the transmit power is further based on the second indication.

30. A method for wireless communication at a network node, comprising: receiving, for a user equipment (UE), an indication of available energy at the UE for a period of time and a buffer size at the UE; generating, for the UE and based on the available energy and the buffer size at the UE, a scheduling decision including a target duty cycle for the UE; and transmitting, for the UE and in accordance with the scheduling decision, a sequence of one or more scheduling grants indicating resources for the UE to use in transmitting the communications in the period of time.